

Evangelos Smith, Ph.D.

evangelos.smith@outlook.com ♦ +1 (936) 217-3987 ♦ [linkedin.com/evangelos-smith](https://www.linkedin.com/in/evangelos-smith)

[evangelossmith.github.io](https://github.com/evangelossmith) ♦ New York

Summary

Researcher with five years of combined computational and experimental experience in heterogeneous thermocatalysis, specializing in catalyst design, reaction kinetics studies, and performance optimization for transformations within the chemicals industry.

Education

University of Wisconsin–Madison

Madison, WI, USA

Ph.D. in Chemical Engineering, Surface Science and Catalysis

April 2026

Thesis: Towards Bridging Materials and Pressure Gaps in Heterogeneous Catalysis

The University of Texas at Austin

Austin, TX, USA

Bachelor of Science in Chemical Engineering, *High Honors* (GPA 3.97/4.00)

May 2021

Experience

Schrödinger, Inc.

New York, NY, USA

Senior Scientist, Materials Science Discovery

May 2026 - Present

University of Wisconsin–Madison

Madison, WI, USA

Ph.D. Candidate with Prof. Manos Mavrikakis

Sept. 2021 - May 2026

Research Focus: Catalytic Sensors for Emissions Abatement & Reaction Monitoring

- Developed liquid crystal (LC)-based sensors for detecting ppm-level NO_x using density functional theory (DFT)
- Conceptualized and developed Pd-supported LC sensors for optically quantifying kinetics and selectivity *in situ* during the hydrogenation of industrially relevant functional groups (nitriles, nitroarenes, and alkynes)
- Synthesized oxide-supported catalysts and conducted reaction kinetics studies, operating continuous-flow reactors to generate experimental benchmarking datasets for microkinetic models
- Applied machine-learned force fields (MLFFs) based on DFT data to accelerate screening and analyze structure–property relationships in catalytic materials

Research Focus: Catalyst Dynamics under Industrial Conditions

- Investigated structural changes of iron ammonia synthesis catalysts using molecular dynamics, DFT, and kinetic modeling
- Identified relationships between adsorbate-induced magnetization changes and catalytic activity to guide promoter screening

Norwegian University of Science and Technology

Trondheim, Norway

Exchange Ph.D. Researcher

May 2023 - Dec. 2024

Research Focus: Biomass Hydrogenation for Sustainable Fuels & Plastics

- Designed, built, and operated high-pressure flow reactors for kinetic studies on the hydrogenation of biomass-derived monomers, quantifying product yields using GC-FID and GC-MS
- Applied DFT calculations to identify activity descriptors that guided rational screening of improved hydrogenation catalysts

Evonik Industries

Lafayette, IN, USA

Additive Manufacturing Research Assistant

Jun. 2021 - Jul. 2021

- Formulated PA12-based polymer composites for selective laser sintering (SLS) of medical implants
- Operated HP Multi Jet Fusion 3D printers and managed full workflow from CAD design to part production

Evonik Industries

Lafayette, IN, USA

Plant Engineering Intern

Jun. 2020 - Jul. 2020

- Supported engineering operations at a GMP-compliant pharmaceutical manufacturing facility, gaining direct exposure to cleanroom practices, documentation standards, and quality protocols
- Established a maintenance schedule for pressure relief devices that decreased maintenance costs by \$73,000 annually

Publications

1st author works denoted by *

- *1. (Sensor Development) **Probing NO₂ Reactivity on Coinage Metal Surfaces through Liquid Crystal Orientational Responses**, [E. Smith](#), H. Yu, H. A. Zhang, T. J. Wolter, R. J. Twieg, N. L. Abbott, M. Mavrikakis, *J. Phys. Chem C* **130**, 9048 (2026). [\[DOI\]](#)
- *2. (Biomass Catalysis) **Selective hydrogenation of furfural acetone over a Cu catalyst: Combined theoretical and experimental study**, M. Rebarchik[†], [E. Smith](#)[†], H. Chang, J. A. Dumesic, M. Mavrikakis, *ACS Catal.* **15**, 6662 (2025). [†]Equal contribution. [\[DOI\]](#)
- *3. (Ammonia Catalysis) **Assessing the performance of exchange-correlation density functionals in describing the iron-catalyzed ammonia synthesis system**, [E. Smith](#), M. Figueras Valls, M. Mavrikakis, *ChemistryEurope* **3**, e202500030 (2025). [\[DOI\]](#)
- *4. (Sensor Development) **First-principles study of O₂ activation over Pd adclusters and embedded clusters in Pd-Au bimetallic surfaces**, [E. Smith](#), T. J. Wolter, R. J. Twieg, N. L. Abbott, M. Mavrikakis, *J. Phys. Chem. C* **129**, 5010 (2025). [\[DOI\]](#)
5. (Surface Characterization) **NO₂ adsorption on oxygen-modified Ag at ambient conditions**, A. Posada-Borbón, T. Wolter, H. Yu, [E. Smith](#), J. J. Schauer, R. C. Van Lehn, V. M. Zavala, N. L. Abbott, M. Mavrikakis, *J. Am. Chem. Soc.* **147**, 43139 (2025). [\[DOI\]](#)
6. (Surface Characterization) **CO-induced roughening of Cu(111): Formation and detection of reactive nanoclusters on metal surfaces**, K. Papanikolaou, L. Xu, [E. Smith](#), M. Mavrikakis, *Surf. Sci.* **754**, 122665 (2025). [\[DOI\]](#) (4th Surface Science Young Investigator Best Paper Award)
7. (Sensor Development) **Synthesis and examination of alkylcyanobiphenyl mesogens with a single fluorine atom at specific locations in the tail**, K. Agrahari, [E. Smith](#), N. L. Abbott, M. Mavrikakis, J. D. Mighion, R. J. Twieg, *Liq. Cryst.* **52**, 976 (2025). [\[DOI\]](#)
8. (Sensor Development) **Racemic and enantiomeric alkoxybiphenyls bearing terminal vicinal fluorine substituents: Synthesis and mesogenic behavior**, K. Agrahari, [E. Smith](#), M. Mavrikakis, J. D. Mighion, R. J. Twieg, *Crystals* **53**, 976 (2025). [\[DOI\]](#)
9. (Surface Characterization) **Moiré patterns in Pt overlayers on gold: A graph neural network interatomic potential study**, T. Wolter, [E. Smith](#), R. Schimmenti, J. Schiøtz, K. W. Jacobsen, M. Mavrikakis, *J. Phys. Chem. C* **128**, 19357 (2024). [\[DOI\]](#)
10. (Sensor Development) **Actuating Liquid Crystals Rapidly and Reversibly by using Chemical Catalysis**, H. Yu[†], J. I. Gold[†], T. J. Wolter, [E. Smith](#), R. J. Twieg, M. Mavrikakis, N. L. Abbott, *Adv. Mater.* **36**, 2309605 (2024). [†]Equal contribution. [\[DOI\]](#)
11. (Sensor Development) **Synthesis and examination of alkoxybiphenyl mesogens with a single fluorine atom at specific locations in the tail**, K. Agrahari, T. J. Wolter, [E. Smith](#), N. L. Abbott, M. Mavrikakis, J. D. Mighion, R. J. Twieg, *Liq. Cryst.* **51**, 1496 (2024). [\[DOI\]](#)
12. (Sensor Development) **Gas-phase microactuation using kinetically controlled surface states of ultrathin catalytic sheets**, N. Bao, Q. Liu, M. Figueras, [E. Smith](#), W. Wang, M. C. Cao, D. A. Muller, M. Mavrikakis, I. Cohen, P. L. McEuen, N. L. Abbott, *Proc. Natl. Acad. Sci.* **120**, e2221740120 (2023). [\[DOI\]](#)

Submitted Manuscripts

1. (Biomass Catalysis) **Perimeter power: Unveiling the role of Ni-TiO₂ interface sites in enhancing acetic acid ketonization catalysis**, P. Tingelstad, [E. Smith](#), N. N. Corneliussen, P. Gramazio, R. L. Peters, K. Rajendran, A. Miró i Rovira, T. Skrzydło, W. M. Tucho, M. Mavrikakis, D. Chen. Preprint at [\[Research Square\]](#)

Selected Presentations

1. **Invited Lecture: Dynamic Roughening of Iron Surfaces under Operando Haber-Bosch Conditions**, PPG Seminar, Madison, WI, Dec. 2025.

2. *In Situ Formation of Fe Clusters During Haber-Bosch Ammonia Synthesis*, 16th European Congress on Catalysis (EuropaCat), Trondheim, Norway, Sept. 2025.
3. *Optical Monitoring of Surface-Catalyzed Reactions Using Chemoresponsive Liquid Crystals*, University of Wisconsin–Madison CATHEX/Olaf. A. Hougen Symposium, Madison, WI, Jun. 2025.
4. *Selective Hydrogenation of Biomass-derived Furans Over Cu Catalysts: Experiments and Theory*, 29th North American Catalysis Society Meeting, Atlanta, GA, Jun. 2025. (*Kokes travel award*)
5. **Invited Lecture:** *Selective Hydrogenation of Biomass-derived Furans Over Cu Catalysts: Experiments and Theory*, 2025 Catalysis Club of Chicago Spring Symposium, Chicago, IL, May 2025.
6. *A First-Principles Study on O₂ Activation on Pd Adclusters and Embedded Clusters in Pd-Au Bimetallic Catalysts*, 2024 AIChE Annual Meeting, San Diego, CA, Oct. 2024.
7. *Reactive Liquid Crystals for the Design of Chemoresponsive Hydrogen Sensors*, Norwegian University of Science and Technology CATHEX/iCSI Catalysis Seminar, Trondheim, Norway, Jun. 2023.

Awards

- **3M RISE Selected Participant** chosen to present doctoral research at 3M’s RISE (Research, Innovation, Service, and Education) Symposium (Sept. 2025)
- **Kokes Travel Award** for presenting at the 29th North American Catalysis Society Meeting (NAM29) (March 2025)
- **Best Presentation Award** for the 2025 Catalysis Club of Chicago Young Scientist Symposium (Jan. 2025)
- **Bird–Stewart–Lightfoot Wisconsin Distinguished Graduate Fellowship** from UW–Madison recognizing graduate students for research achievements (Dec. 2024)
- **4th Surface Science Young Investigator Best Paper Award** for “CO–induced roughening of Cu(111): Formation and detection of reactive nanoclusters on metal surfaces” (Oct. 2024)
- (3) **Roland A. Ragatz Teaching Assistant Awards** from UW–Madison recognizing excellence in undergraduate instruction (Dec. 2022–Dec 2023)
- **INEOS Olefins & Polymers Senior Plant Design Project Award** for a 2nd place submission for a proposed naptha reforming facility (Aug. 2020)

Teaching Experience

University of Wisconsin–Madison

Madison, WI, USA

- *Undergraduate Mentor, Special Problems (CBE 599)*: Advised an undergraduate student’s research project on the dehydrogenation of biomass-derived monomers over Cu catalysts (Jan. 2024 - May 2025)
- *Teaching Assistant, (1) Kinetics and Catalysis; (2) Thermodynamics of Mixtures; (3) Chemical Process Thermodynamics* (Sept. 2022 - Dec. 2023)

The University of Texas at Austin

Austin, TX, USA

- *Teaching Assistant, Chemical Process Safety (CHE 364S)* (Aug. 2020 - May 2021)

Gus Garcia Young Men’s Leadership Academy

Austin, TX, USA

- *Club Coach*: Developed and presented lesson plans for a 6th grade science course as part of The University of Texas’ UTeach Outreach program (Jan. 2020 - May 2020)

Skills

Computational: DFT (VASP, Gaussian, QE), Python, MATLAB, LAMMPS, Machine-learned force fields (MLFFs)

Laboratory Continuous-flow reactors, catalyst synthesis/characterization, GC-FID, GC-MS, BET, CO chemisorption

Languages: English, Greek

Work Authorization (No Sponsor): United States of America, European Union